MIL-F-49183(ER)
12 September 1978

MILITARY SPECIFICATION

FILTER, DIRECT CURRENT, POWER F-1461( )/VRC

1. SCOPE

1.1 Scope. This specification covers a power line ripple filter intended for use with Controls, Intercommunication Set C-2297/VRC and C-2298/VRC in some tracked vehicle installations. The filter is designated as Filter, Direct Current, Power F-1461( )/VRC. (See 6.1 and 6.8).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to U.S. Army Electronics Research and Development Command, ATTN: DELET-DP, Fort Monmouth, N.J. 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC-5915

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

NN-P-71	Pallet, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
PPP-B-20	Bag, Cotton, Mailing
PPP-T-76	Tape, Pressure Sensitive, Adhesive Paper, Water Resistant
PPP-T-97	Tape, Packaging, Industrial, Filament Reinforced
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock Corner
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple Wall
PPP-S-760	Strapping, Non-metallic (and Connectors)
PPP-C-843	Cushioning, Material, Cellulosic
PPP-C-1797	Cushioning Material, Resilient, Low Density, Unicellular Polyproplylene Foam
QQ-S-781	Strapping, Steel, Flat and Seals

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Military

MIL-P-116	Preservation-Packaging, Methods of
MIL-S-901	Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment and Systems, Requirement for
MIL-I-6869	Impregnants for Aluminum and Magnesium
MIL-P-11268	Parts, Materials and Processes Used In Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection By Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized Unit Loads
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-276	Impregnation of Porous Nonferrous Metal Castings
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-965	Parts Control Program
MIL-STD-1188	Commercial Packaging of Supplies and Equipment
MIL-STD-1562	Lists of Standard Microcircuits

DRAWINGS

Electronics Command

DL-SC-B-951290	Filter, Direct Current, Power F-1461( )/VRC

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Construction. The equipment shall be constructed in accordance with drawing and Data List DL-SC-B-951290 and the requirements of this specification.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample units for first article inspection and approval. (See 4.3).

3.3 Parts, materials and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268 including the selection requirement therein shall apply, except that MIL-STD-965 shall be used in lieu of MIL-STD-749 for parts control procedures.

3.4 Finish. The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings. (See 4.4)

3.5 Marking.

3.5.1 General. Marking shall conform to Specification MIL-M-13231. (See 4.4). Front panel marking shall be Group I as described in that specification.

3.5.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.3 Serial numbers. Filter, Direct Current,Power F-1461( )/VRC shall be serial numbered.

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3.6 Die castings. (See 4.4)

3.6.1 Quality of die castings. Quality of die castings shall be as follows:

a. Surface condition shall be clean, dry and free from chips.

b. Cast surface finish shall be of a painting grade which allows some streaks and chill marks which can be covered and hidden by paint.

c. Flash must be removed by grinding or filing.

d. The casting, after impregnation, shall be air tight, and no leakage shall be allowed under a pressure of 10 psi gage.

e. Flatness shall be as specified in the government furnished drawings.

f. Dimensions shall be as specified in the government furnished drawings.

3.6.2 Impregnation of aluminum die castings. All aluminum sand and die castings shall be impregnated in accordance with Standard MIL-STD-276, using impregnant material Type II per Specification MIL-I-6869. Castings which have been impregnated shall be marked "IMP". The impregnation shall be to correct general seepage only and shall not be used to correct poor foundry techniques or significant porosity (see 4.4).

3.7 Environmental service conditions. The equipment shall meet the following environmental service conditions;where a test is referenced, meeting the test shall be considered as compliance with the requirement.

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3.7.1 Temperature. The equipment shall meet the following temperature requirements. Measurements shall be taken with the equipment operating with 25.5 ±1 volt input.

3.7.1.1 Low temperature.

a. Operating. The equipment shall be operable without degradation in specified performance at ambient air temperatures as low as -50°F. During testing as specified in 4.10.1.1, the equipment shall meet the electrical requirements of 3.9.

b. Storage and transportation. The equipment shall withstand exposure to ambient air temperatures as low as -70°F.

3.7.1.2 High temperature.

a. Operating. The equipment shall be operable without degradation in specified performance at ambient air temperatures as high as +125°F plus the effects of solar radiation at a rate of 360 BTU/ft^2/hr. During testing as specified in 4.10.1.2, the equipment shall meet the electrical requirements of 3.9.

b. Storage and transportation. The equipment shall withstand exposure to ambient air temperatures as high as +160°F.

3.7.2 Humidity. The equipment shall exhibit no physical damage such as corrosion, rust, blistering, swelling or deterioration of parts and materials. During testing as specified in 4.10.2, the equipment shall meet the electrical requirements of 3.9.

3.7.3 Altitude. The equipment shall be operable without degradation in specified performance at altitudes up to 15,000 feet above sea level and shall withstand air transportation at altitudes up to 40,000 feet above sea level. The equipment shall meet the electrical requirements of 3.9, both during and after testing as specified in 4.10.3

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3.7.4 Vibration and shock.

3.7.4.1 Vibration. The equipment shall withstand vibration and shock induced during vehicular transport as part of a mobile assemblage or installed within or upon a military vehicle, over all types of roads and cross-country terrain; and shocks encountered during servicing. Equipment to be installed within or upon an armored vehicle shall be resistant to high level shock caused by ballistic impact, air explosions, and similar combat field conditions. After being tested in accordance with 4.10.4 there shall be no physical damage and the equipment shall meet the electrical requirements of 3.9.

3.7.4.2 Shock, ballistic. The equipment shall be capable of meeting the requirements of Table I.

Table I - Shock

Paragraph Heading	Paragraph	Performance After Test
Shock, Ballistic	4.10.5	Operable (Note A). Any physical damage shall be minor only, as classified in MIL-STD-252

NOTE A: The equipment shall meet the Electrical Requirements of 3.9.

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3.7.5 Immersion. The equipment shall be submersion proof to a covering depth of three (3) feet of water for not less than two (2) hours with no evidence of leakage. (See 4.10.6)

3.7.6 Fungus. After the test of 4.10.7 the filter shall be visually examined, using a 10-power magnifier. The filter shall show no more than six minute unrelated spots, each no greater than 0.015 square inch in area, of sparse microbial growth (see 6.4) as evidenced by growth colonization (see 6.4) (which includes branching (see 6.4) and sporulation (see 6.4)) on or within each cubic foot or fraction thereof, of equipment assembly volume. Isolated instances of partial tubular germination (see 6.4) shall not be included in this evaluation.

3.7.7 Salt fog. After the salt fog test of 4.10.8, the filter, when examined visually with the aid of a 10-power magnifier, shall show no evidence of degradation, such as flaking, pitting, blistering or loosening of finish on metal surfaces; or exfoliation (see 6.4) of metal. Electrical operation is not required.

3.8 Preconditioning. (See 4.6)

3.8.1 Bounce preconditioning. The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6.

3.9 Electrical requirements for F-1461( )/VRC. The F-1461( )/VRC is designed to provide ON-OFF control and power line filtering for the Controls, Intercommunication Set C-2297/VRC and C-2298/VRC in some tracked vehicle installations.

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3.9.1 Continuity. Electrical continuity of wiring and of other connections shall be verified by a point-to-point test (see 4.9.1). Circuits K and V shall be individually shielded and the shields grounded.

3.9.2 Zener overvoltage protection. The output voltage of the F-1461( )/VRC shall be less than 36 Vdc ±5% when a voltage of 50 Vdc is applied at the input connector. (See 4.9.2).

3.9.3 Operating functions. The F-1461( )/VRC shall provide the following operating functions:

3.9.3.1 Power ON-OFF control by operation of circuit breaker handle. (See 4.9.3).

3.9.3.2 Power ON indication. (See 4.9.4).

3.10 Interchangeability. Like unit assemblies, subassemblies and replaceable parts shall conform to Requirement 7 of MIL-STD-454. (See 4.11)

3.11 Air-seal test. When tested as specified in 4.7, the decrease in vacuum shall not exceed 0.01 pounds per square inch guage (PSIG), per minute.

3.12 Safety engineering.

3.12.1 Personnel hazard. Personnel hazards shall be kept to a minimum through compliance with Requirement 1 of MIL-STD-454.

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3.12.2 Edge rounding. All exposed edges and corners shall be rounded.

3.12.3 Radioactive materials. Radioactive materials shall not be used.

3.13 Workmanship. The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

a. In MIL-P-11268:

General requirements for plastic material and plastic parts.
Wiring and cabling, including:
- Slack.
- Protection.
- Splicing.
- Connections, general.
- Grounding, general.
- Shielding on wire and cable.

Soldering and brazing.
Cleaning of units.
General requirements for securing of parts.
Controls.
Self-tapping screws.

b. In this specification:

3.4	Finish,protective
3.5	Marking
3.6	Die castings

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First Article Inspection (does not include packaging). (See 4.3)

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before packaging. (See 4.5)

(2) Quality conformance inspection of packaging. (See 4.14)

4.3 First article. Unless otherwise specified in the contract, the first article inspection shall be performed by the contractor.

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4.3.1 First article units. The contractor shall furnish four (4) first article units of the complete Filter, Direct Current,Power F-1461( )/VRC.

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, preconditioning specified in 4.6, and the inspections specified for Group A, Group B and Group C (see Tables III, IV and V respectively). The inspection shall be performed in the following order, 4.4, 4.6, Group A and Group B for all units, and Group C as specified in Table II. After completion of the Group C environmental tests, conforming units shall be reinspected and shall pass all Group A inspection.

4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract.

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Table II - Order Of Environmental Tests

Inspection	NUMBER OF SAMPLES Unit 1	Unit 2	Unit 3	Unit 4
Non-damaging				
Temperature				
Low				1
High				2
Altitude				3
Potentially damaging				
Vibration	1			
Shock, Ballistic	2			
Humidity		1		
Fungus			1	
Immersion	3			
Salt Fog				4

NOTE: The order on non-damaging tests and the choice of which units are subjected to the non-damaging tests may be varied if convenient, except that the vibration test should be performed on the same unit used for the shock test.

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4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before packaging:

Item	Where Required
Parts, Materials and Processes Used In Electronic Equipment	3.3
Finish	3.4
Marking	3.5
Casting Impregnation and Quality	3.6

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspection specified in 4.4, 4.6 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract shall be inspected for conformance to the inspection specified in Table III. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table III.

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Table III - Group A Inspection

Inspection	Req. Para.	Insp Para.	AQL Major	AQL Minor
Visual and Mechanical	3.13	4.13	1.0%	4.0%
Electrical				
Continuity	3.9.1	4.9.1	1.0% for group combined	N/A
Zener Overvoltage Protection	3.9.2	4.9.2		
Power ON-OFF Control	3.9.3.1	4.9.3		
Power ON indication	3.9.3.2	4.9.4		
Air Seal	3.11	4.7	1%	N/A

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government.

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4.5.2 Group B inspection. Group B inspection shall normally be performed in inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table IV and to the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plan. The Group B sampling plans for the AQL's listed in Table IV shall be inspection level S-4.

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in an order satisfactory to the Government.

TABLE IV - Group B Inspection

INSPECTION	REQT PARA	INSP PARA	AQL
Interchangeability	3.10	4.11	6.5%
Immersion	3.7.5	4.10.6	10.0%

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table V. Samples shall be selected in accordance with 4.5.3.1.

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TABLE V - Group C Inspection

INSPECTION	REQT PARA	INSP PARA
Subgroup I		
Immersion	3.7.5	4.10.6
Subgroup II		
Temperature	3.7.1	4.10.1
Humidity	3.7.1	4.10.2
Altitude	3.7.3	4.10.3
Vibration	3.7.4.1	4.10.4
Shock, ballistic	3.7.4.2	4.10.5
Subgroup III		
Fungus	3.7.6 (*)	4.10.7
Salt Fog	3.7.7 (*)	4.10.8

*Unless otherwise specified in contract or order a different sample unit shall be used for each of these tests.

4.5.3.1 Sampling for Group C inspection.

4.5.3.1.1 Subgroup I. For this subgroup two units shall be selected from the first 50 production units for the first Group C inspection. For subsequent Group C inspection, two units from each successive 250, or two units per month if less than 250 are produced monthly, shall be selected.

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4.5.3.1.2 Subgroup II. For this subgroup, two units shall be selected from the first 50 production units for the first Group C inspection. For subsequent Group C inspections, two units every 6 months shall be selected.

4.5.3.1.3 Subgroup III. For this subgroup one unit of each type control shall be selected from the first production lot for the first Group C inspection.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in an order satisfactory to the Government.

4.5.3.3 Group C failures. Actions required relative to Group C failures shall be specified in the contract.

4.5.4 Reinspection of conforming Group B and C sample units. Unless otherwise specified, sample units which have been subjected to and passed both Group B and Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.6 Bounce preconditioning. The unit, with shock mounts (if any) removed or blocked, shall be placed on its cover on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneatales, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to inspection. (See 3.8).

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4.7 Air-seal test. The equipment shall be opened and closed again in such manner as to break and remake the seal. Immediately thereafter, the equipment as field transported shall be subjected to a vacuum of 1 pound per square inch (1 pound per square inch less than the atmospheric pressure surrounding the equipment) applied to the interior of the equipment inclosure. The vacuum then shall be valved-off and the interior pressure measured during the ensuing period of 1 minute. During this 1-minute period the decrease in vacuum shall not exceed 0.01 pound per square inch. The gage used for measurement of the vacuum shall be of such accuracy that a differnce of 0.01 pound can be determined readily. Control lamp cover shall be removed to provide connection for air seal test F-1461( )/VRC. (See 3.11).

4.8 Standard test conditions. Unless otherwise specified, all tests shall be conducted at 25.5 volts dc input and room ambient conditions of temperature and humidity.

4.9 Electrical tests (See 3.9).

4.9.1 Continuity. (See 3.9.1). Without any other connections and with all switches in any position, measure continuity between all corresponding pins of J-71 and J-72 with the exception of pin C. There shall also be an open circuit from pin A to any other pin of either receptacle with switches in any position.

4.9.2 Zener overvoltage protection. (See 3.9.2). Connect 50 volts dc between pin C (+) through a 270 ohm 1 watt resistor and pin A (ground) of the input receptacle J-72. The open circuit voltage measured between pin C (+) and pin A (ground) of the output receptacle J-71 shall be less than 38 volts dc.

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4.9.3 Power On-Off control. Connect a 270 ohm 4 watt resistance load between pin C and pin A of the output receptacle J-71. Connect 25.5 volts dc between pin C (+) and pin A (ground) of the input receptacle J-72. Monitor voltage across resistor with a suitable meter and operate circuit breaker handle on and off to test its operation. Circuit breaker handle up position shall be power ON.

4.9.4 Power ON indication. While performing test of paragraph 4.9.3 observe panel lamp for proper operation.

4.10 Environmental service conditions tests. Environmental service conditions tests shall be performed as detailed below. Each unit subjected to these tests shall have passed Group A tests and shall be resubmitted for retest after completion of all environmental service tests.

4.10.1 Temperature.

4.10.1.1 Low temperature. The equipment shall be tested in accordance with Method 502.1, Procedure I, of MIL-STD-810. The storage temperature shall be -70°F, and the operating temperature shall be -50°F. Prior to testing the equipment shall meet full specification performance for the measurements specified in 3.7.1.1.

4.10.1.2 High temperature. The equipment shall be tested in accordance with Method 501.1, Procedure II, of MIL-STD-810. For Step 7, the chamber temperature shall be adjusted to +160°F for test item operation (this includes the effects of solar radiation). Prior to testing the equipment shall meet full specification performance for the measurements specified in 3.7.1.2.

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4.10.2 Humidity. The equipment shall be tested in accordance with Method 507.1, Procedure II, of MIL-STD-810. Prior to testing the equipment shall meet full specification performance for the measurements specified in 3.7.2. In Step 6 measurements shall be taken during the last five (5) hours of the last cycle. Do not move equipment from the humidity chamber for measurements and do not leave power on after measurements have been completed. If any equipment fails to meet the specified performance during cycling it shall be realigned or readjusted once and the test continued. No repair or replacement of parts shall be permitted. If the equipment still fails to meet the specified performance during cycling or fails subsequently, it does not pass the test. In addition, if the equipment fails to meet specified requirements after final conditioning and adjustment it does not pass the test. The test shall be performed with the cover off.

4.10.3 Altitude. The equipment shall be tested in accordance with Method 500.1, Procedure I, of MIL-STD-810. Prior to testing the equipment shall meet full specification performance for the measurements specified in 3.7.3.

4.10.4 Vibration. The equipment shall be subjected to the test of Method 514.2, Procedure VIII, using Curve W, Figure 514.2-6, of MIL-STD-810. Test time shall be three (3) hours in each of three (3) mutually perpendicular directions. At the conclusion of the vibration test, the filter shall be visually inspected for any evidence of mechanical damage and operated to verify satisfactory performance in relation to pre-test performance data. (See 3.7.4.1).

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4.10.5 Shock, ballistic.

a. Test method. The test shall be conducted on the "Shock Testing Machine for Light Weight Equipment" shown in Specification MIL-S-901. The filter shall be secured in its normal operating position to test plate fixture 4a, figure 5 of MIL-S-901, by means of the same fasteners used for vehicular installation of the control. The test shall consist of a total of 9 blows: one each 1 foot blow, 3 foot blow, and 5 foot blow on the back, side, and top of the test plate. As an alternative to reorienting the plate for the blows on the side of the plate, equivalent rotation of the control under test is permissible. (See 3.7.4.2).

b. Performance. Upon completion of the test and before repairs are made, tests shall be made for compliance with the following test: Paragraph 4.9 (Electrical tests).

4.10.6 Immersion (See 3.7.5). The equipment, as prepared for field transportation shall be tested in accordance with Method 512.1, Procedure I, of MIL-STD-810.

4.10.7 Fungus. The equipment shall be subjected to the fungus test specified in Method 508.1, Procedure I, of MIL-STD-810. There shall be abundant growth colonization (6.4) on 50 percent or more of the area of the control item after 14 and 28 days. No cleaning of the equipment is permitted for 72 hours prior to this fungus test. Handling prior to and during testing, shall be accomplished without contamination of the equipment. Inability of the equipment to meet the requirements of 3.7.6 shall constitute failure of this test.

4.10.8 Salt fog. The equipment shall be subjected to the salt fog test specified in Method 509.1 Procedure I, of MIL-STD-810. No corrosion products shall be deposited on the test facility. Inability of the equipment to meet the requirements of 3.7.7 shall constitute failure of this test.

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4.11 Inspection for interchangeability. The points listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.10. When a listed dimension is not within specified or design limits it shall be considered a major defect.

a. Case. The maximum overall outside dimensions and location of four mounting screw holes.

b. Bottom cover. The maximum overall outside dimension, length and width, and location of four mounting screws.

4.12 Safety engineering. An inspection shall be performed to verify compliance with those portions of 3.12 which can be determined visually.

4.13 Visual and mechanical inspection. The equipment shall be examined for the defects listed in Standard MIL-STD-252.

4.14 Quality conformance inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

4.15 Rough handling test (packaging). When rough handling test in accordance with MIL-P-116 is required by the contract (see 6.2), the power ON-OFF control test (see 4.9.3) shall be conducted to determine freedom from operational malfunction caused by rough handling.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or commercial, as specified (see 6.2c.).

5.1.1 Level A.

5.1.1.1 Cleaning. Each filter shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each filter shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservative application. No preservative required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Hardware. Each set of hardware, including spares, shall be stowed in the bag conforming to PPP-B-20. Close the bag with the drawstring.

5.1.1.4.2 Spare lamp. Cushion each lamp with cushioning conforming to PPP-C-1797. Secure the cushioning with tape conforming to PPP-T-76.

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5.1.1.4.3 Filter, direct current, power F-1461( )/VRC. Each filter shall be unit packed method III. Cushion the filter together with the hardware and lamp, specified in 5.1.1.4.1 and 5.1.1.4.2, with cushioning conforming to PPP-C-1797. Secure cushioning with tape conforming to PPP-T-76. Place the cushioned unit within a close-fitting box conforming to PPP-B-636, W5c. Closure shall be in accordance with the appendix of the box specification.

5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1.

5.1.3 Commercial preservation. Preservation shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B or commercial, as specified (see 6.2). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of Filter, Direct Current, Power F-1461( )/VRC, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

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5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers, packed as specified in 5.2.1 except that container strapping may be omitted, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.2 Level B. A quantity of Filter, Direct Current, Power F-1461( )/VRC, unit packed as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film. When a single unit is shipped to a single destination no packing is required; the unit container shall serve as the shipping container. Reinforcing shall be as specified above.

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5.2.2.1 Unitization. Palletization shall be required when quantities per destination comprise a pallet load. A qualtity of Filter, Direct Current, Power F-1461( )/VRC, unit packed as specified in 5.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.3 Commercial packing. Packing shall be in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-1188.

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6. NOTES

6.1 Intended use. The Filter, Direct Current, Power F-1461( )/VRC is used with Control,Intercommunication Set C-2297( )/VRC or C-2298( )/VRC to provide supply voltage filtering. (see 1.1)

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Type required.

c. Level A, level B or commercial preservation and packing (see Section 5).

d. First Article pack(s) as follows:

Makeup of pack(s).
Number of each kind of pack to be submitted.
Inspection to be performed thereon.

e. Marking and shipping of samples.

f. Place of final inspection.

g. Technical literature required, test tools and quantity of running spare parts.

h. When rough handling test is required. (see 4.15)

i. Size data. Size limitations cited herein are based on the dimensions of the equipment for which these requirements were established. Requests to exceed these limitations must be supported by data indicating why limitations cannot be met and recommend a revised limitation.

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6.3 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.4 Definitions.

6.4.1 Equipment divisions. The terms used herein for equipment divisions conform to MIL-STD-280.

6.4.2 Definitions of inspection terms used herein conform to MIL-STD-109. The following are abbreviated definitions selected from those given therein for immediate reference.

6.4.2.1 Inspection. Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.4.2.2 Examination. Examination consists of simple, generally nondestructive determinations of compliance, without use of special testing equipment.

6.4.2.3 Testing. Testing consists of determinations of compliance, using technical means.

6.4.3 Exfoliation. Exfoliation is corrosion along the grain boundaries of the metal resulting in the peeling or separating, or both, of successive layers of the metal. The appearance resembles loose book pages or onion skin peeling.

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6.4.4 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots on secondary items growing from the main stem or filament (hypha).

6.4.5 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

6.4.6 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms, when present in large numbers, may provide a colony visible to the naked eye.

6.4.7 Sporulation. Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

6.4.8 Tubular germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungae filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.5 Humidity. Natural climatic extremes may include 100-percent relative humidity with condensation up to 4 hours at a time. This requirement (see 3.7.2) is specified by MIL-STD-210. The humidity test (see 4.10.2) using MIL-STD-810, is a procedure that gives a practical test approximating the natural conditions.

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6.6 Group C inspection. Approval to ship may be withheld, at the discretion of the Government pending the decision from the contracting officer on the adequacy of corrective action.

6.7 Location of air-seal test. It is desirable that the air-seal test (4.7) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. Any preparation for shipment which would require breaking of the equipment seal should be accomplished prior to the air-seal test so that the seal may remain intact thereafter. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.8 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design: for example: F-1461W/VRC. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (see 1.1)

6.9 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE MIL-F-49183 (ER) FILTER, DIRECT CURRENT, POWER F-1461( )/VRC

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

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